

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS: 50

Annexure A
CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

*I **Kartheeka Repalle (192372289)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Kartheeka Repalle

Annexure A
CONSTRAINT-BASED PROBLEM SOLVING

1. A financial institution must secure all web transactions using TLS, but legacy systems only support older SSL versions. Design a solution to ensure secure communication.

Constraints:

- A. Backward compatibility must be maintained for 20% legacy clients
 - B. Sensitive data must not be transmitted using SSL
 - C. System downtime must be minimal during migration
 - D. Cost of implementation should be optimized
-
2. An e-commerce platform is considering implementing the SET protocol for payment security but currently uses TLS-based encryption. Analyze and propose a decision.

Constraints:

- A. Must support high transaction throughput (10,000 transactions/minute)
 - B. Compatibility with modern browsers is required
 - C. Implementation complexity should be minimized
 - D. Regulatory compliance (PCI-DSS) must be ensured
-
3. A corporate network is experiencing repeated malware infections that spread rapidly without user interaction. Propose a containment and prevention strategy.

Constraints:

- A. The network consists of 500+ interconnected systems
 - B. Downtime per system should not exceed 10 minutes
 - C. Detection must occur within 2 minutes of infection
 - D. Solution must differentiate between virus, worm, and Trojan behavior
-
4. An organization needs to deploy a firewall architecture to protect its internal network from external threats and insider attacks. Design the solution.

Constraints:

- A. Must support both packet filtering and application-level inspection
- B. Latency introduced should not exceed 5 ms per request
- C. Should handle at least 1 Gbps traffic
- D. Must allow secure remote access for employees

5. A large enterprise network generates excessive false positives in its IDS, leading to alert fatigue. Propose an optimized IDS/IPS solution.

Constraints:

- A. Detection accuracy must be at least 95%
- B. False positive rate should be reduced below 5%
- C. System must process real-time traffic of 2 Gbps
- D. Solution should integrate with existing firewall infrastructure

1. COMPARISON OF AH AND ESP IN IPSEC

ANSWER:

Authentication Header (AH) and Encapsulating Security Payload (ESP) are two protocols used in IPsec to secure network communication.

AH provides data integrity, authentication, and protection against replay attacks. However, it does not provide confidentiality because the original data is not encrypted.

ESP provides confidentiality through encryption and can also provide data integrity, authentication, and replay protection. Therefore, ESP offers more complete security services than AH.

Comparison:

Feature	AH	ESP

Confidentiality	No	Yes
Integrity	Yes	Yes
Authentication	Yes	Yes
Encryption	No	Yes
Replay Protection	Yes	Yes
Overhead	Lower	Higher
VPN Suitability	Limited	High

AH has lower processing overhead because it does not encrypt data.

ESP requires more processing because encryption and integrity protection are performed. Modern VPN hardware can reduce this performance overhead.

AH may be suitable when only authentication and integrity are required. ESP is more suitable for confidential business communication, financial information, employee traffic, and VPN connections.

For a VPN deployment, ESP is generally preferred because it provides both confidentiality and integrity. ESP can therefore protect sensitive information from being read or modified while travelling through the public Internet.

Conclusion:

ESP is more appropriate for most modern VPN deployments because it provides stronger overall protection and supports encryption, integrity, authentication, and replay protection.

2. COMPARISON OF TRANSPORT MODE AND TUNNEL MODE IN IPSEC

ANSWER:

IPSec provides two main operating modes: Transport Mode and Tunnel Mode. Both protect network communication but differ in the amount of the IP packet that is protected.

Comparison:

Feature	Transport Mode	Tunnel Mode

Protection	Payload	Entire IP packet
New IP Header	No	Yes
Overhead	Lower	Higher
Host-to-Host	Suitable	Possible

Network-to-Network	Less suitable	Highly suitable
VPN Usage	Limited	Common

Transport mode protects the payload of the original IP packet while the original IP header remains visible. It is mainly suitable for host-to-host communication where two individual systems communicate directly.

Advantages of Transport Mode:

- Lower processing overhead.
- Lower packet overhead.
- Suitable for direct host-to-host communication.
- Efficient when end systems provide IPSec protection.

Disadvantages:

- Original IP addressing information remains visible.
- Less suitable for connecting complete networks.
- Provides less network-level privacy than tunnel mode.

Tunnel mode protects the complete original IP packet and places it inside a new IP packet. It is commonly used for site-to-site VPN connections.

Advantages of Tunnel Mode:

- Protects the complete original IP packet.
- Hides internal IP addresses from the public network.
- Suitable for network-to-network communication.
- Commonly used in corporate VPNs.

Disadvantages:

- Higher packet overhead.
- Requires additional processing.
- May slightly reduce network efficiency.

Conclusion:

Transport mode is more suitable for host-to-host communication, while tunnel mode is more suitable for network-to-network communication. For corporate site-to-site VPNs, tunnel mode is generally preferred because it provides stronger protection for internal network traffic.

3. COMPARISON OF PGP AND S/MIME

ANSWER:

PGP and S/MIME are technologies used to provide secure email communication. Both provide encryption, integrity, authentication, and digital signatures, but their key-management and trust models are different.

Comparison:

Feature	PGP	S/MIME

Key Management	User controlled	Certificate based
Trust Model	Web of Trust	CA/PKI
Scalability	Moderate	High
Enterprise Use	Moderate	High
Central Management	Limited	Strong
Integration	Depends on software	Strong
Encryption	Yes	Yes
Digital Signature	Yes	Yes

PGP uses public-key cryptography and commonly follows a decentralized Web of Trust model. Users can create their own keys and establish trust with other users.

Advantages of PGP:

- Flexible key management.
- User-controlled trust.
- Strong encryption.
- Supports digital signatures.
- Useful in decentralized environments.

Disadvantages of PGP:

- Key management can become difficult.
- Trust relationships can be complicated.
- Centralized administration is more difficult.
- Large organizations may face challenges managing many keys.

S/MIME uses public-key cryptography together with digital certificates issued by Certificate Authorities. It is commonly integrated with enterprise email and PKI systems.

Advantages of S/MIME:

- Centralized certificate management.
- Strong enterprise integration.
- Structured trust model.
- Easier identity management.
- Supports certificate renewal and revocation.

Disadvantages of S/MIME:

- Requires PKI infrastructure.
- Certificate management is required.
- Initial deployment may require additional administration.

Recommendation:

For a corporate email system, S/MIME is generally more suitable. It provides centralized certificate management, structured trust, and good integration with enterprise email systems.

PGP may be preferred when users require decentralized key management and greater control over their own trust relationships.

Conclusion:

S/MIME is generally recommended for large corporate environments because it provides better scalability, centralized management, and enterprise integration.

4. RULE-BASED VS MACHINE LEARNING-BASED EMAIL SPAM DETECTION

ANSWER:

Rule-based and machine learning-based techniques are commonly used to identify spam and phishing emails. Rule-based systems depend on predefined conditions, while machine learning systems learn patterns from previously classified email data.

Comparison:

Feature	Rule-Based	Machine Learning

Method	Fixed rules	Learned patterns
Setup	Simple	More complex
Adaptability	Low	High
New Spam Detection	Limited	Better
Maintenance	Manual	Training required
False Positives	Can be high	Can be reduced
Scalability	Moderate	High
Processing	Fast	Model dependent

Rule-Based Detection:

Rule-based systems use predefined rules to identify suspicious emails. Examples include detecting spam keywords, suspicious

senders, unusual domains, excessive links, or specific attachment types.

Advantages:

- Easy to implement.
- Easy to understand.
- Fast processing.
- Rules can be directly controlled.
- Suitable for simple spam patterns.

Disadvantages:

- Requires frequent rule updates.
- New spam patterns may bypass the rules.
- Attackers can modify message content.
- Too many rules may increase false positives.

Machine Learning-Based Detection:

Machine learning systems learn patterns from previously classified spam and legitimate emails. They can analyze email headers, message content, URLs, sender information, attachments, and other features.

Advantages:

- Can identify complex patterns.
- Better adaptability to changing spam.
- Can process large amounts of email.
- Can reduce dependence on manually created rules.
- Can improve detection of previously unseen patterns.

Disadvantages:

- Requires suitable training data.
- Requires continuous monitoring and retraining.
- Incorrect training data can reduce accuracy.

- More complex to implement and maintain.

False Positive Handling:

False positives occur when legitimate emails are incorrectly classified as spam. To reduce this problem, organizations can use different confidence levels.

High-confidence spam can be blocked or quarantined. Medium-confidence messages can be marked as suspicious, while low-confidence messages can be delivered normally.

A hybrid approach combining rule-based and machine learning techniques can provide better protection.

Conclusion:

Rule-based detection is simple, fast, and easy to manage, while machine learning provides better adaptability to evolving spam patterns. A combination of both methods is generally more effective for modern email security.

5. MANUAL VS AUTOMATED IPSEC KEY MANAGEMENT

ANSWER:

IPSec requires cryptographic keys and Security Associations (SAs) to secure network communication. Key management can be performed manually or automatically using protocols such as Internet Key Exchange (IKE).

Comparison:

Feature	Manual Management	Automated IKE

Key Generation	Manual	Automatic

Scalability	Low	High
Administration	Difficult	Easier
Error Risk	High	Lower
Key Renewal	Manual	Automatic
Security	Moderate	Stronger
Large Enterprise	Difficult	Suitable
Operational Cost	High	Lower

Manual Key Management:

In manual key management, administrators configure keys and IPSec security parameters directly on network devices.

Advantages:

- Simple for small networks.
- Easy to understand.
- Does not require an automated key-management system.

Disadvantages:

- Difficult to manage many devices.
- Keys must be changed manually.
- High possibility of configuration errors.
- Difficult to replace compromised keys quickly.
- Poor scalability.
- Increased administrative workload.

Automated Key Management Using IKE:

IKE automatically negotiates security parameters and establishes cryptographic key material between IPSec peers. It also supports automatic Security Association creation and renewal.

Advantages:

- Automatic key generation.

- Automatic key renewal.
- Better scalability.
- Reduced configuration errors.
- Easier management of large VPN environments.
- Supports secure authentication and negotiation.
- Reduces administrative workload.

Impact on Security:

Automated key management reduces the risk of expired, reused, or incorrectly configured keys. Regular key renewal also improves the security of IPSec communication.

Impact on Scalability:

Manual key management becomes increasingly difficult as the number of VPN devices increases. IKE allows devices to negotiate keys automatically, making it much more suitable for large organizations.

Operational Complexity:

Manual management may be simple initially, but it becomes difficult and time-consuming as the network grows. Automated management requires initial configuration but significantly reduces long-term administrative work.

Conclusion:

Manual key management may be acceptable for small laboratory or test networks. For large organizations, automated key management using IKE is preferred because it provides better security, scalability, reliability, and operational efficiency.